Koneru Lakshmaiah Education Foundation
(Deemed to be University estd. u/s. 3 of the UGC Act, 1956)
Off-Campus: Bachupally-Gandimaisamma Road, Bowrampet, Hyderabad, Telangana - 500 043.
Phone No: 7815926816, www.klh.edu.in

DEPARTMENT OF CSE

PROJECT ABSTRACT SUBMISSION

COURSE: SOA PROGRAMMING AND MICROSERVICES– (24SDCS03)
A.Y. 2026 - 2027

TITLE OF THE PROJECT		
S. No.	**University ID**	**Name**
1	2420030349	A.Srinivas
2	2420030562	M.Pranay
3	2420030562	S.Roshan
4		
Name of the Guide		**Venkateswari Ketineni**

Abstract:

The rapid growth of e-commerce has significantly transformed the way customers discover, purchase, and manage furniture products online. Traditional furniture shopping often requires customers to visit multiple stores to compare products, prices, and availability, while conventional online platforms may face challenges related to scalability, security, service reliability, and efficient management of customers, sellers, products, carts, orders, and payments. These challenges highlight the need for a flexible and scalable online furniture marketplace that can provide a seamless shopping experience while supporting independent sellers and efficient backend operations.

To address this need, the proposed project, **"Oakora: A Scalable Microservices-Based Online Furniture Marketplace,"** introduces a distributed e-commerce platform that utilizes modern web technologies and a microservices architecture. The system is designed to provide customers with an integrated platform for discovering furniture, managing their accounts, adding products to carts, placing orders, and completing payments, while allowing sellers to manage their products and monitor their business activities.

The proposed system introduces a modular microservices architecture in which individual business functionalities are developed and deployed as independent services. The system consists of dedicated services for user management, seller management, product management, cart management, order processing, and payment handling. **Spring Boot** is used for developing the backend microservices, while **Spring Cloud Eureka** provides service discovery and **Spring Cloud Gateway** acts as a centralized entry point for routing client requests to the appropriate services. **OpenFeign** facilitates communication between microservices, while **MySQL** databases are maintained independently for individual services following the database-per-service principle.

The user management service provides secure registration, authentication, profile management, and role-based access for customers, sellers, and administrators. **The product and seller services** enable sellers to add, update, delete, and manage furniture products while allowing customers to browse, search, and filter products based on categories and other attributes. **The cart and order services** manage shopping-cart operations, order placement, order history, and order-status tracking. **The payment service** provides a controlled payment-processing workflow, while a notification service can be integrated to provide updates regarding orders and transactions.

The frontend application is developed using **React and Tailwind CSS**, providing a responsive and user-friendly interface for customers and sellers. The platform follows a clear separation between presentation, business logic, and data management, enabling individual services to be developed, tested, maintained, and scaled independently. **JWT-based authentication and Spring Security** are incorporated to protect sensitive user and service operations.

Collectively, the proposed Oakora platform demonstrates how microservices architecture can be applied to develop a scalable, modular, and secure online furniture marketplace. By separating core business functionalities into independent services, the system improves maintainability, service isolation, scalability, and flexibility compared with a tightly coupled monolithic architecture. The integration of modern backend technologies, distributed service management, secure authentication, and a responsive frontend provides a strong foundation for a reliable digital furniture marketplace and offers potential for future enhancements such

as cloud deployment, recommendation systems, real-time notifications, analytics, and automated CI/CD pipelines.

Project Architecture/Project Overview:

